I aim to create **multimodal foundation models** that are **useful for and generalize to diverse users** across linguistic, cultural, and geographic contexts. I am excited about the following key areas:
a) Advancing **multimodal alignment** and language grounding in foundation models to foster inclusivity, by developing **scalable data curation** techniques for reflecting real-world diversity.
b) Improving **agentic abilities**, by endowing models with better **reasoning**, **retrieval** and **in-context learning** abilities that enable generalization in unseen environments. This is crucial because foundation models are deployed in novel settings while interacting with users from diverse backgrounds.
I worked towards these goals at **Google DeepMind** with Dr. Partha Talukdar, Dr. Sriram Ganapathy and Dr. Trevor Cohn, **FAIR Labs Meta** with Dr. Pengchuan Zhang, and during my undergraduate thesis and long-term collaboration with **IIT Delhi**, **IBM Research**, and **USC** with Prof. Parag Singla, and Prof. Xiang Ren. Below, I outline my previous and ongoing research, its impact, and the future directions I aim to pursue.

Data Curation for Inclusive Multimodal Models

Foundation models struggle to generalize to niche domains due to the scarcity of training data. For example, consider tail languages: over 60% of Indic languages with more than 100,000 speakers have almost zero web-text presence, highlighting the challenges of data curation for these languages. This scarcity, compounded by the fact that key cultural and linguistic concepts often reside in non-text modalities, necessitates improved multimodal alignment and cross-modal knowledge transfer. My past work addresses this by developing scalable algorithms for a) Modifying large-scale multimodal data for obtaining better alignment signals b) Leveraging synthetic data for cross-modal transfer.

During my AI residency at Meta, I addressed cross-modal alignment deficiencies in contrastively trained VLMs, specifically their limitations in **compositional reasoning**, through a data curation approach. I realized that the contrastive training objective of VLMs introduces inconsistencies by matching images with texts of varying complexity. My "coarse-to-fine" data curation technique converted image captions into scene graphs, decomposed into sub-graphs of varying complexity, including hard negatives. Inspired by supervised contrastive learning, I developed a novel loss for matching images to multiple text scene-graphs, significantly improving compositionality. This work [1] was published in **EMNLP 2023** (and received an **Oral** acceptance at ICCV and EMNLP workshops).

Given the proven utility and cost-effectiveness of synthetic data, at DeepMind, I proposed using **synthetic data** for instruction tuning of multimodal (speech-text) models for cross-lingual and cross-modal alignment. I explored some of these ideas as a core contributor to a large effort aimed at improving the multilingual capabilities of **Gemini** via synthetic data curated from multimodal sources. These projects provided valuable experience in tackling challenges in cross-modal alignment, and collaborating with large teams across geographies, ultimately contributing to significant improvements in Gemini's multilingual performance. I also learned **large-scale engineering** for training models with **billions of parameters**.

High-quality evaluations are crucial for advancing language models in niche domains, yet challenging, due to scarcity of web data and human-annotators. For this, at DeepMind, I led the development of IndicGenBench [2], a large-scale generative benchmark for evaluating LLMs on Indic languages. My experiments revealed that in-context transfer learning performs better when transferring from a high-resource Indic language (like Hindi) than from English. I also observed significant performance disparities between languages due to biased tokenizers (trained primarily on English), challenges with polysemy (words with multiple meanings) and code-mixing (mixing languages within a sentence). Open-source IndicGenBench [HuggingFace, Press Coverage: *Indian Express, Google Blog*] is contributing to state-of-the-art Indic LLM development (e.g., Sarvam AI) and was published at **ACL 2024**. This project provided firsthand experience in LLM evaluation research and also surfaced the **need for more efficient evaluation strategies** beyond expensive human-annotated benchmarks.

To address this need, I explored two key directions. First, I investigated generating synthetic evaluation data, combined with lightweight human verification to ensure quality. This significantly reduced the cost while I developed a multilingual SpeechQA dataset. Second, I proposed efficient metrics such as compression ratio, to "measure" tokenized data directly, and showed them to be correlated with downstream performance. This reduced the need for human-annotated downstream benchmarks, and enabled faster iteration for improving data quality (under review at ICASSP [3]).

My contributions to benchmarks and synthetic data curation are actively being used to improve Gemini, enhancing its multilinguality and impacting millions of users.

Inference Time Generalization and Reasoning

While better data curation techniques create more capable static foundation models, it does not provide the ability to **generalize "in the wild"** to novel and out-of-distribution scenarios, for example while interacting with a user to solve a hard coding task, or reason about a complicated cultural phenomenon. For better generalization, I aim to make progress on multiple fronts – a) improvements to in-context learning (ICL) to help adapt models to novel situations, b) better reasoning and deliberate thinking to enable better responses on hard and out-of-distribution cases, and c) progress in retrieval to resolve knowledge gaps for effective generalization.

My recent project aims at evaluating and improving ICL in long-context multimodal foundation models. While ICL using long-context models has shown promise in simple text domains, my experiments show that multimodal ICL performance does not scale well with many demonstrations, exhibiting degradation with large context lengths. I am also evaluating if models can learn from weakly-supervised data and different modalities like speech in the context, essential for true generalization in data-sparse domains. Additionally, I found that retrieval augmentation with long-context models provides significant gains to knowledge intensive tasks. I believe the next step is to incorporate reasoning in the retrieval process for generating better queries for retrieval.

My interest in generalization and reasoning arose during my first research projects while pursuing my undergraduate. With Prof. Parag Singla, I developed a modular approach to instruction-guided image editing, enhancing performance on complex compositional tasks using weak supervision from VQA tasks [5] (published in **EMNLP 2023**) and recently helped design a planning based method for compositional text-to-image generation, achieving significant gains over SoTA T2I models [6]. With Prof. Xiang Ren at InkLab USC, I developed a modular model that improved out-of-distribution generalization in reasoning tasks, published in **ACL 2022** [4].

My experiences reinforce the importance of deliberate reasoning for generalization to complex and out-of-distribution problems at test-time. Additionally, our understanding of ICL in real-world settings is primitive and improvements require explicit interventions at training time via data or modeling changes.

Future Interests

Through my experiences I believe that building generalizable models requires a) scalable knowledge transfer across modalities, especially for niche domains where traditional data annotation is prohibitive and b) techniques that promote test time generalization.

Therefore, I am interested in advancing **modality alignment** in large multimodal models. I aim to develop techniques for effectively extracting knowledge, particularly in the form of synthetic data, from non-text modalities like speech. This has the potential to enrich text data, improving LLMs in domains where multimodal knowledge and language grounding is essential. For example, this synthetically curated data can encompass certain dialectal nuances, cultural concepts, behavioral cues, and other information not readily available in text alone. Additionally, creating better multimodal models requires addressing the current limitations in cross-modal alignment. I would like to explore fine-grained modality alignment objectives during training, including better loss functions and better aligned multimodal training data. This can have various applications – as an example, aligning speech and text in high-resource settings, offers a promising avenue for knowledge transfer to low-resource languages lacking sufficient text data.

I believe **data curation** methodologies are essential for improving modern day foundation models as well as for training agents. I would like to fundamentally understand and improve data selection techniques for large-scale pre-training and fine-tuning, design algorithms for curating better synthetic data, establish best practices in combining real and synthetic data, and understand the tradeoffs involved while using these techniques with respect to various downstream tasks. Additionally, I would like to explore how multimodal signals can guide better data curation techniques.

Beyond building static models, I am excited about the real world generalization capabilities of foundation models. This can be effectively studied by building useful **multimodal agents** for the real world. This direction is quite fascinating since it provides a perfect setting for assessing real-world reasoning and generalization capabilities. Building agents that effectively assist users and navigate diverse environments requires crucial advancements in, and synchronization of, **reasoning**, **retrieval** and **in-context learning**. I believe agents must effectively leverage long multimodal contexts, perform robust retrieval to resolve knowledge gaps, and engage in enhanced reasoning to navigate diverse, real-world environments and unseen user interactions. For improving training of multimodal agents, I would like to explore techniques for generating synthetic multimodal agent trajectories and investigating how agents can learn from these experiences to generalize and make better decisions. Additionally, improving fine-grained visual reasoning skills is essential for digital agents that interact with people and other agents through visual and audio signals. My diverse experiences in multimodality and reasoning would allow me to effectively make advances in this domain.

Why PhD at UC Berkeley? I aspire to be a long-term academic and a principal investigator, and UC Berkeley offers the ideal environment for pursuing my doctoral studies. There are several faculty members whose work is well matched with my research goals. I am interested in working with **Prof. Sewon Min** because of our closely-aligned interests in building models that better generalize at test time to long-tail concepts. While my current research uses retrieval with long-context models to better generalize to niche concepts, I am very excited about Prof. Sewon's recent work on non-parametric models, which allows better generalization via a completely different paradigm. I would like to work with **Prof Gopala Krishna Anumanchipalli** on problems in speech and multimodal learning, and how we can leverage speech to better learn about non-text phenomena such as dysfluency. With **Prof. Kurt Keutzer** and **Prof. Dawn Song**, I would like to work on improving multimodal agents, and also explore how to better use inference time compute for improving reasoning capabilities of these agents. I would also like to make inference time sampling and search techniques more efficient, contrary to the general consensus that we can use as much compute as we want during inference, and for this, Prof Keutzer's

expertise would be invaluable. Additionally, **Prof. Dan Klein**'s work on using multimodal signals for language learning is well-matched with my interests in cross-modal transfer and grounded multimodal learning and I would love to have a chance to work with him.

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